AGRONOMIC INTELLIGENCE BRIEF

1 Acre Integrated Cropping System — Full Agronomic Breakdown (Adamawa)

Permanent zobo + chili border with rotational / relay core — rainfed productivity, income continuity, soil resilience.

FIELD 63.4 × 63.4 M | ≈ 1 ACRE · 4,020 M² | BORDER 2 M | 3 SYSTEM OPTIONS

TOTAL FIELD STRUCTURE

Total field size

63.4 m × 63.4 m ≈ 4,020 m² — ≈ 43,264 sq ft — ≈ 1 acre

PERMANENT BORDER (ALL OPTIONS)

Pattern

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Zb Ch Zb Ch Zb Ch
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Zb Zobo Ch Chili — productive first and second season.

BORDER DIMENSION

Effective width: 2 m

Inner block: 59.4 m × 59.4 m

LAND USE RATIO

COMPONENT	AREA	%
Zobo border	246 m²	6.1%
Chili border	246 m²	6.1%
Inner production	3,528 m²	87.8%

BORDER CROPS — FULL AGRONOMY

Zobo (Hibiscus)

Area ≈ 246 m² · **Spacing** 1×1 m · **Plants** ≈246

Seed 0.8–1.2 kg · **Yield** 250–450 kg dry calyx

Biomass stems (fiber / fuel / mulch) · **Harvest** Aug–Nov

Chili pepper

Area ≈246 m² · **Spacing** 0.6×0.45 m · **Plants** ≈900

Nursery seed 25–40 g · **Yield** 2–4 t fresh

Harvest Aug–Dec

OPTION 1: EGUSI + DROP CORN → SESAME OR COWPEA RELAY SYSTEM

1 ACRE INTEGRATED CROPPING SYSTEM WITH PERMANENT ZOBO + CHILI BORDER

ADAMAWA, NIGERIA

FIRST SEASON (MAY – JULY)

- Egusi as main crop
- Drop corn for support, food security and staking
- Zobo + Chili border remains continuous



LAYOUT: Alternating Border Strips – Zobo (Zb) and Chili (Ch) 2 Rows Deep Around Entire Field

FIRST SEASON (MAY – JULY)

Crop	Area (m²)	Spacing	Population	Seed Required	Expected Yield
Zobo	246	1m x 1m	246 plants	0.8 – 1.2 kg	250 – 450 kg dry calyx
Chili	246	0.6m x 0.45m	≈ 900 plants	25 – 40 g	2 – 4 tons fresh
Egusi	3,528	1m x 1m	≈ 3,500 plants	3 – 4 kg	500 – 900 kg seed
Drop Corn	3,528	2m x 0.75m	≈ 2,000 – 2,500 stands	5 – 6 kg	500 – 900 kg grain

ROLES IN FIRST SEASON

- **Egusi** – main crop, ground cover & moisture conservation
- **Drop corn** – support, staking, food security
- **Zobo + Chili border** – pest barrier, windbreak, continuous income

SECOND SEASON (AUGUST – NOVEMBER)

- Egusi harvested
- Relay crop planted into residual moisture
- Zobo + Chili border continues to produce
- Optional drop sorghum for late season grain



SECOND SEASON (AUGUST – NOVEMBER) – RELAY OPTIONS

OPTION A: SESAME RELAY (with optional drop sorghum)

Crop	Area (m²)	Spacing	Population	Seed Required	Expected Yield
Sesame	3,528	0.75m x 0.15m	≈ 31,000 – 35,000 plants	1.5 – 2.5 kg	350 – 650 kg
Sorghum (Optional)	3,528	2m x 0.75m	≈ 1,500 – 2,000 stands	2 – 3 kg	300 – 500 kg grain

OPTION B: COWPEA RELAY (with optional drop sorghum)

Crop	Area (m²)	Spacing	Population	Seed Required	Expected Yield
Cowpea	3,528	0.75m x 0.2m	≈ 20,000 – 24,000 plants	20 – 25 kg	500 – 850 kg grain
Sorghum (Optional)	3,528	2m x 0.75m	≈ 1,500 – 2,000 stands	2 – 3 kg	300 – 500 kg grain

OPTION 1: SUMMARY (PER ACRE)

Permanent Border (Year Round)
- **Zobo:** 246 plants → 250 – 450 kg dry calyx
- **Chili:** ≈ 900 plants → 2 – 4 tons fresh

First Season (May – July)
- **Egusi:** ≈ 3,500 plants → 500 – 900 kg seed
- **Drop Corn:** ≈ 2,000 – 2,500 stands → 500 – 900 kg grain

Second Season (Aug – Nov)
- **Option A (Sesame):** 350 – 650 kg seed
- **Option B (Cowpea):** 500 – 850 kg grain
- **Optional Sorghum:** 300 – 500 kg grain

KEY BENEFITS OF OPTION 1

- Continuous income from Zobo + Chili
- High land productivity with multiple harvests
- Covers soil all year, reduces erosion
- Diversified outputs – food, oilseed, fibre, spice
- Strong resilience to dry spells

CROPPING CALENDAR (OPTION 1)



Option 1 — Egusi + drop corn → sesame or cowpea relay; permanent zobo + chili border.

First season (May–July)

Inner block ≈3,528 m²

Egusi

1×1 m · ≈3,500 stands · seed 3–4 kg · yield 500–900 kg seed — cover, weeds, moisture.

Drop corn

Scattered 2×0.75 m equiv. · 2,000–2,500 stands · 5–6 kg seed · 500–900 kg grain — food security, support, wind.

Second season (Aug–Nov)

BORDER	POP.	YIELD
Zobo	246	250–450 kg
Chili	900	2–4 t

Relay A — sesame

3,528 m² · 0.75×0.15 m · 31k–35k plants · 1.5–2.5 kg seed · 350–650 kg — moisture, drought, oilseed.

Optional drop sorghum

1,500–2,000 stands · 2–3 kg · 300–500 kg — drought insurance.

Relay B — cowpea

0.75×0.2 m · 20k–24k · 20–25 kg seed · 500–850 kg — N fixation, fodder.

Option 1 — total system output

CROP	POPULATION	SEED	YIELD
Zobo	246	1 kg	250–450 kg
Chili	900	30 g	2–4 t
Egusi	3,500	3–4 kg	500–900 kg
Drop corn	2,000–2,500	5–6 kg	500–900 kg
Sesame relay	31k–35k	2 kg	350–650 kg
OR cowpea	20k–24k	20–25 kg	500–850 kg
Opt. sorghum	1,500–2,000	2–3 kg	300–500 kg

OPTION 2: GROUNDNUT → RELAY SESAME (WITH ZOBO + CHILI BORDER)

1 ACRE INTEGRATED CROPPING SYSTEM

FIRST SEASON (JUNE – AUGUST)

Main crop:
Groundnut

Border (Permanent):
Zobo + Chili

Field size:
1 Acre (63.4m x 63.4m)

Border width:
2m (all around)

Inner crop area:
3,528 m² (87.8%)

Border area:
492 m² (12.2%)



SECOND SEASON (AUGUST – NOVEMBER)

Relay crop:
Sesame

Border (Permanent):
Zobo + Chili continue

Note:
Groundnut harvested in August. Sesame is established using residual moisture.



FIRST SEASON (JUNE – AUGUST)

Crop	Spacing	Area (m²)	Population	Seed Requirement	Expected Yield
Zobo (Hibiscus)	1m x 1m	246	246 plants	0.8 – 1.2 kg	250 – 450 kg dry calyx
Chili Pepper	0.6m x 0.45m	246	≈ 900 plants	25 – 40 g (nursery)	2 – 4 tons fresh
Groundnut	0.5m x 0.2m	3,528	≈ 176,000 plants	70 – 90 kg (shelled seed)	1.2 – 1.8 tons pods

KEY BENEFITS – FIRST SEASON

- Groundnut fixes nitrogen and improves soil fertility
- Provides soil cover and suppresses weeds
- Zobo + Chili border reduces pest entry and wind speed
- Multiple income sources from border crops

SECOND SEASON (AUGUST – NOVEMBER)

Crop	Spacing	Area (m²)	Population	Seed Requirement	Expected Yield
Zobo (Hibiscus)	1m x 1m	246	246 plants	-	250 – 450 kg dry calyx
Chili Pepper	0.6m x 0.45m	246	≈ 900 plants	-	2 – 4 tons fresh
Sesame	0.75m x 0.15m	3,528	≈ 31,000 – 35,000 plants	1.5 – 2.5 kg	350 – 650 kg seed

KEY BENEFITS – SECOND SEASON

- Sesame utilizes residual moisture and provides oilseed income
- Continuous border production ensures cash flow
- Improves soil structure and reduces erosion

SUMMARY – OPTION 2 (PER ACRE)

Permanent Border Area	Inner Crop Area	Total Main Season Yield (Groundnut + Border)	Total Second Season Yield (Sesame + Border)	Total Annual Outcome
492 m² (12.2%)	3,528 m² (87.8%)	≈ 1.45 – 2.25 tons pods + calyx + chili	≈ 0.85 – 1.10 tons seed + calyx + chili	High productivity with continuous income

NOTE: All population figures are approximate. Yields depend on rainfall, soil fertility, and management practices.

AGRONOMIC ADVANTAGES

- Soil Fertility Improvement
- Drought Tolerance
- Weed Suppression
- Pest Management
- Multiple Income Streams

Option 2 — Groundnut → relay sesame; zobo + chili border.

First season (Jun–Aug)

BORDER	POP.	YIELD
Zobo	246	250–450 kg
Chili	900	2–4 t

Groundnut

3,528 m² · 0.5×0.2 m · ≈176,000 plants · 70–90 kg shelled seed · 1.2–1.8 t pods — N fixation, cover, weeds.

Second season (Aug–Nov)

BORDER	POP.	YIELD
Zobo	246	250–450 kg
Chili	900	2–4 t

Sesame relay

0.75×0.15 m · ≈31,000 · 2 kg seed · 350–650 kg — residual moisture, dry-season fit.

Option 2 — total system output

CROP	POPULATION	SEED	YIELD
Zobo	246	1 kg	250–450 kg
Chili	900	30 g	2–4 t
Groundnut	176,000	70–90 kg	1.2–1.8 t
Sesame	31,000	2 kg	350–650 kg

OPTION 3: SOYBEAN + DROP CORN OR SORGHUM (ONE SEASON SYSTEM)

1 ACRE INTEGRATED CROPPING SYSTEM WITH PERMANENT ZOBO + CHILI BORDER

ONE SEASON (JUNE – OCTOBER)

Main crop:
Soybean

Intercrop (choose one):
Drop Corn
OR
Drop Sorghum

Border (Permanent):
Zobo + Chili

Field size:
1 Acre (63.4m x 63.4m)

Border width:
2m (all around)

Inner crop area:
3,528 m² (87.8%)

Border area:
492 m² (12.2%)



ONE SEASON (JUNE – OCTOBER)

Crop	Spacing / Pattern	Area (m²)	Population	Seed Requirement	Expected Yield
Zobo (Hibiscus)	1m x 1m	246	246 plants	0.8 – 1.2 kg	250 – 450 kg dry calyx
Chili Pepper	0.6m x 0.45m	246	≈ 900 plants	25 – 40 g (nursery)	2 – 4 tons fresh
Soybean	0.5m x 0.1m	3,528	≈ 300,000 – 350,000 plants	45 – 60 kg	1.3 – 2.0 tons grain
CHOOSE ONE INTERCROP					
	1 stand every 2m (eq. 2m x 0.75m)	3,528	≈ 2,000 stands	4 – 5 kg	400 – 800 kg grain
OR					
	1 stand every 2m (eq. 2m x 0.75m)	3,528	≈ 1,500 – 2,000 stands	3 – 4 kg	400 – 700 kg grain

KEY BENEFITS – OPTION 3

- Soybean fixes nitrogen and improves soil fertility
- Drop corn or sorghum provides food security and structural support
- Zobo and chili border reduces pest entry and wind speed
- One-season system – lower cost and labour requirement
- High net return with balanced risk under Adamawa conditions

AGRONOMIC ADVANTAGES

Nitrogen Fixation | Soil Fertility Improvement | Wind Break | Pest Management | Drought Tolerance | Profitable & Low Risk

OPTION 3A: SOYBEAN + DROP CORN

OR

OPTION 3B: SOYBEAN + DROP SORGHUM

SUMMARY – OPTION 3 (PER ACRE)

PERMANENT BORDER (BOTH SEASONS)					MAIN SEASON (JUNE – OCTOBER)					TOTAL SYSTEM OUTPUT (ONE SEASON)
Crop	Area (m²)	Population	Seed Req.	Expected Yield	Crop	Area (m²)	Population	Seed Req.	Expected Yield	
Zobo (Hibiscus)	246	246 plants	0.8 – 1.2 kg	250 – 450 kg dry calyx	Soybean	3,528	≈ 300,000 – 350,000 plants	45 – 60 kg	1.3 – 2.0 tons grain	• Total border area: 492 m² (12.2%)
Chili Pepper	246	≈ 900 plants	25 – 40 g (nursery)	2 – 4 tons fresh	Drop Corn (Option 3A)	3,528	≈ 2,000 stands	4 – 5 kg	400 – 800 kg grain	• Total inner crop area: 3,528 m² (87.8%)
					OR					• Expected total food grain (soybean + cereal): 1.7 – 2.8 tons (combined)
					Drop Sorghum (Option 3B)	3,528	≈ 1,500 – 2,000 stands	3 – 4 kg	400 – 700 kg grain	• Continuous income from border crops throughout the season

CROP CALENDAR – OPTION 3 (ONE SEASON)

Activity	May	June	July	Aug	Sept	Oct	Nov
Land Preparation	■						
Planting (Soybean + Drop Corn/Sorghum)	■						
Weeding		■	■	■			
Border Management (Zobo + Chili)	■	■	■	■	■	■	■
Harvest (Soybean)					■		
Harvest (Corn/Sorghum)						■	
Zobo & Chili Harvest			■	■	■	■	■

Note: All population figures are approximate. Actual yield depends on rainfall, soil fertility, variety, and management practices.

Option 3 — Soybean + drop corn or sorghum (one season); zobo + chili border.

Season (Jun–Oct)

BORDER	POP.	YIELD
Zobo	246	250–450 kg
Chili	900	2–4 t

Soybean

0.5×0.1 m · 300k–350k plants · 45–60 kg seed · 1.3–2 t grain — N fixation, mechanization, soil regen.

Drop corn

2×0.75 m equiv. · ≈2,000 stands · 4–5 kg seed · 400–800 kg — food security, structure.

Option 3 — total output

CROP	POP.	SEED	YIELD
Zobo	246	1 kg	250–450 kg
Chili	900	30 g	2–4 t
Soybean	300k–350k	45–60 kg	1.3–2 t
Drop corn	2,000	4–5 kg	400–800 kg

SYSTEM-WIDE COMPARISON

PARAMETER	OPT 1	OPT 2	OPT 3
Production intensity	Very high	Moderate	High
Labor	Very high	Moderate	Lower
Soil improvement	Moderate	Very high	High
Drought tolerance	High	Very high	High
Income diversity	Very high	Moderate	Moderate
Mechanization	Low	Moderate	High
Risk	Medium	Low	Low–med
Harvest flow	Excellent	Excellent	Good

FINAL INTERPRETATION

Not a standard intercrop. A **permanent border + rotational core**: zobo + chili as continuous perimeter; core rotates with rains; relays use residual moisture; drop cereals buffer drought and food security; economically active across the Adamawa rainfed cycle.